The Security Tax

Pricing Bitcoin's Physical Vulnerability
from Insurance Market Data

The case for Great Wall

Yuri da Silva Villas Boas · April 2026

Abstract

Physical attacks on Bitcoin holders ("$5 wrench attacks") are growing at 43% annually, while the individual-whale population grows at only 24%. The community's consensus defense — obscurity — is both the dominant advice and a structural impediment to adoption. Rather than fitting a discount to Bitcoin price model residuals, this paper estimates the security cost from first principles by cross-referencing wrench attack rates with insurance premiums for comparable portable luxury goods. Restricted to individual (non-institutional) holders with >$100K in BTC, the severity-adjusted annual security cost reaches 0.64% of holdings at the 2026 annualised pace — already inside the gold-storage-fee range. Because Bitcoin's value incorporates speculative expectations and the attack trend is worsening, the forward-looking discount exceeds the current cost: 3.6% over a 5-year horizon and 8.2% over 10 years at a 10% discount rate. Great Wall closes this gap through Tacit Knowledge-Based Authentication (TKBA) — memory-held entropy encoded onto a user-specific perturbation of the Burning Ship fractal, gated by a calibrated Argon2 derivation, with optional outsourcing of a time-lock puzzle to an anonymous Lightning Network marketplace. The project is entirely free software (MIT or Apache-2.0).

The Trend

On 12 February 2026, the account `@BitcoinNewsCom` posted on X:

> ***ATTACKS ON BITCOINERS IN FRANCE ACCELERATING*** *— Out of 47 documented attacks against Bitcoin holders in France since 2017, 38 have occurred in just the last 13 months.*

A prominent community account, `@BitcoinersHQ`, replied: *"No one needs to know you own Bitcoin."* The original poster endorsed this as *"Golden advice."*

This exchange reflects an established consensus across the Bitcoin security community:

- Jameson Lopp (Casa CTO): *"The best things Bitcoiners can do to stay safe is to remain private. Don't go around telling anyone about your Bitcoin holdings."*[1]
- Ledger Academy: *"Never share how much bitcoin you own. Keep a low profile online and offline."*[2]
- ChainSec: *"Hide your crypto stickers. Don't wear crypto merch. Blend in completely."*[3]

The reply by `@yurivillasboas` to the thread above exposed the structural contradiction:

> *I'm sorry, but that premise undermines propagation of adoption. Plus, everybody cannot be needles in a haystack. If everyone is needle, there's no hay.* ***Maximalism vs Obscurity. Choose one.***

The community endorses obscurity *in the same threads* where it documents obscurity's failure. Obscurity *is* the security strategy — and it imposes a measurable cost. This paper estimates that cost.

[1] `https://www.theblock.co/post/384018`
[2] `https://www.ledger.com/academy/glossary/wrench-attack`
[3] `https://chainsec.io/wrench-attack/`

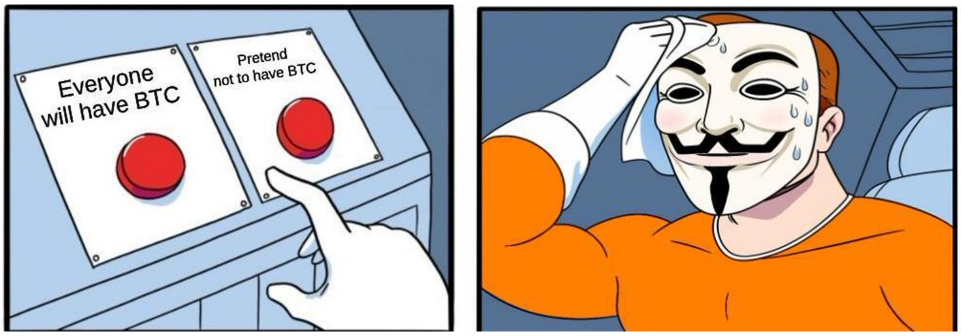

Figure 1: The *Maximalism vs. Obscurity* dilemma, circulated widely among Bitcoin holders. The contradiction the dilemma dramatises is not incidental: it is the community's current physical-security posture.

An Insurance Framework

How insurance markets price physical security risk

Insurance premiums for high-value portable goods — jewelry, luxury watches, collectible cars — directly price the cost of physical vulnerability. This market is deep, mature, and empirically well-studied. The table below compiles premium quotes from six independent sources:

Source / tier	**Premium** (% of value/yr)	**Notes**
Specialty jewelry insurers		
Jewelers Mutual[4]	1.0–2.0%	Dominant US specialty insurer; surcharge 25–50% for items > \$15K
GEICO / Jewelers Mutual[5]	1.0–2.0%	White-label of Jewelers Mutual
General insurers / aggregators		
NerdWallet / III[6]	1.0–2.0%	US national average
Bankrate[7]	1.0–2.0%	Avg. \$13–\$20 per \$1K/yr
Progressive[8]	1.0–2.0%	Via endorsement or standalone
Crime-environment variation		
Low-crime rural (e.g. New Hampshire)[9]	∼1.0%	Burglary ∼48/100K
US national average[10]	1.5%	Burglary 229/100K
High-crime urban (e.g. San Francisco)[6]	3.0–3.5%	Burglary ∼500/100K

The convergence across independent sources is striking: every major insurer and aggregator quotes 1–2% of appraised value at the US national average. The premium varies roughly 3× across the crime spectrum, reflecting the insurer's actuarial estimate of frequency × severity. The **pure premium** (net of administrative loading) is approximately 0.9% at the US average crime rate. This follows from the industry loss ratio: the inland marine line (parent statutory category for personal articles floaters) reports direct loss ratios of 44–49%,[11] implying a pure premium of ∼0.5× the 1.5% midpoint gross premium ≈ 0.75–0.9%.

What Bitcoin lacks

No insurance market exists for self-custody Bitcoin. This is not an oversight — it reflects a fundamental obstacle: a wrench attack can extract the *entire* holdings in a single event with no possibility of recovery, reversal, or tracing. An insurer cannot verify the loss, assess negligence, or recover stolen funds.

[11] AM Best, *Inland Marine Insurance Market Analysis*, 2024. Loss ratio 44–49% post-pandemic.

The absence of insurance means the holder bears the full actuarial risk *plus* an uninsured risk premium — the additional discount that risk-averse agents demand when they cannot offload tail risk. Under standard CRRA utility ($\gamma = 2$–5), the certainty-equivalent cost of a concentrated, total-loss risk is 1.5–3× the actuarial expectation.[12]

Model Construction

Target population

Not all Bitcoin holders are plausible wrench attack targets. We restrict the analysis to **individual (non-institutional) holders with >$100K in BTC**:

- **Included:** individuals holding via hardware wallets, software wallets, paper wallets, or personal exchange accounts — anyone who can be physically coerced into transferring funds.
- **Excluded:** exchange company wallets, Bitcoin ETF custodians (Coinbase Custody, Fidelity), corporate treasuries (MicroStrategy), sovereign funds. These entities have professional security, multi-party authorization, and cannot be compromised by coercing a single individual.

The comparison is individual-to-individual: individual jewelry owners face burglary and robbery; individual Bitcoin whales face wrench attacks. The institutional layer is excluded from both sides.

Whale population estimates are necessarily approximate. They are derived from on-chain address clustering (Glassnode), exchange-reported user demographics, and BTC price (which determines the $100K threshold in BTC terms). We provide sensitivity analysis in Section 8.

The security cost formula

Let $a(t)$ be the number of documented wrench attacks in year t, $W(t)$ the estimated individual whale population (thousands), and m the underreporting multiplier. The **estimated attack rate per 100K whales** is:

$$\rho(t) = \frac{a(t) \times m}{W(t) \times 1{,}000} \times 100{,}000 = \frac{100\, m \cdot a(t)}{W(t)} \tag{1}$$

This is directly comparable to standard crime rates (burglary per 100K, robbery per 100K) used in insurance pricing.

We translate this crime rate into an insurance-equivalent annual cost using the empirical relationship from jewelry insurance:

$$\boxed{c(t) = \underbrace{\sigma}_{\text{severity}} \times \underbrace{\frac{\pi_j}{r_j}}_{\text{cost per unit crime}} \times \rho(t)} \tag{2}$$

where:

- $\pi_j = 0.9\%$ — jewelry pure premium at US average crime rate
- $r_j = 229$ — US average burglary rate per 100K
- $\sigma = 6$ — severity multiplier (derived below)

The ratio $\pi_j / r_j = 0.00393\%$ per unit of crime rate is the **transfer coefficient**: the insurance market's empirical pricing of one unit of per-100K crime rate for portable luxury goods. Multiplying by σ adjusts for the greater severity of Bitcoin attacks relative to jewelry theft.

[12] Pratt, J.W., "Risk Aversion in the Small and in the Large," *Econometrica* 32(1/2), 1964; Eeckhoudt, Gollier & Schlesinger, *Economic and Financial Decisions under Risk*, Princeton, 2005, Ch. 4–5.

Severity multiplier

Bitcoin wrench attacks differ from jewelry theft along two dimensions:

1. **Total vs. partial loss.** A jewelry burglary typically results in partial loss: some items are taken, others overlooked; recovery rates are nonzero. Insurance loss ratios for jewelry average ∼30–50% of coverage. A wrench attack on a Bitcoin holder typically extracts the *entire* wallet balance — the attacker forces a complete transfer. Severity ratio: ∼3×.
2. **Uninsured vs. insured.** Jewelry theft is insurable; the owner pays a premium and offloads tail risk. Bitcoin self-custody risk cannot be insured. Risk-averse agents demand a premium for bearing uninsurable catastrophic risk. Conservative estimate: 2× the actuarial expectation.

Combined: $\sigma = 3 \times 2 = 6$. This is conservative; arguments for $\sigma = 10$–15 (accounting for violence premium, psychological trauma, and irrecoverability) are defensible.

The transfer coefficient

Combining the parameters, the **transfer coefficient** 0.00023580786026200870 maps a per-100K crime rate directly to an annual cost:

$$0.00023580786026200870 = \sigma \times \frac{\pi_j}{r_j} = 6 \times \frac{0.009}{229} = 0.000236$$

All values in this paper are computed live by LaTeX from these parameters. Changing any input (e.g. $\sigma = 10$ instead of 6, or $m = 10$ instead of 3) recomputes the entire analysis.

Data and Results

Input data

Year	**Attacks** $a(t)$	**Indiv. Whales** $W(t)$ (K)	**BTC Year-End** (\$)	**Gold Avg** (\$/oz)	**\$100K threshold** (BTC)
2014	1	10	318	1 266	314
2015	5	15	430	1 160	233
2016	4	50	960	1 251	104
2017	12	500	14 156	1 257	7.1
2018	25	200	3 693	1 269	27
2019	8	250	7 160	1 393	14
2020	14	600	29 000	1 770	3.4
2021	33	1000	46 300	1 799	2.2
2022	23	400	16 500	1 800	6.1
2023	18	700	42 000	1 943	2.4
2024	41	900	93 000	2 386	1.1
2025	70	1000	88 445	3 435	1.1
2026†	28	1100	97 000	∼3 500	1.0

Sources: Attacks from Lopp's GitHub tracker[13]; whale estimates from Glassnode[14] (on-chain clustering), exchange demographics, adjusted for BTC price; BTC year-end from CoinMarketCap; gold from LBMA. Crime rates: FBI UCR 2024 (burglary 229/100K), FBI UCR 2023 (robbery 67/100K).[15]

†2026 data through April 13 (103 of 365 days), pulled from Jameson Lopp's physical-bitcoin-attacks tracker. Annualised pace: 99 attacks — already above the 2025 total. Of the 28 YTD attacks, 20 occurred in France; France has accounted for roughly 70% of documented 2026 incidents.

[13] Lopp, J., *Known Physical Bitcoin Attacks*, `https://github.com/jlopp/physical-bitcoin-attacks`. See also: "Investigating Wrench Attacks," *Advances in Financial Technologies (AFT)*, LIPIcs, 2024.

[14] Glassnode, on-chain entity clustering and address analytics, `https://glassnode.com`.

[15] FBI, *UCR Summary of Reported Crimes in the Nation*, 2024–2025, `https://cde.ucr.cjis.gov/`.

Gray rows (2014–2018) are excluded from the per-capita analysis: whale populations below ∼250K make per-capita rates unreliable (e.g. 5 attacks among 15K whales in 2015 gives 100/100K — above the US robbery rate — but one attack more or fewer swings the estimate by 20%). The per-capita analysis begins at 2019 ($W \geq 250$K).

Annual security cost (baseline: $m = 3$, years 2019–2026)

Year	Attacks $\times m$	Rate/100K	Ann. cost $c(t)$	Ann. cost
2019	24	9.6	0.002264	0.226%
2020	42	7	0.001651	0.165%
2021	99	9.9	0.002334	0.233%
2022	69	17.2	0.004068	0.407%
2023	54	7.7	0.001819	0.182%
2024	123	13.7	0.003223	0.322%
2025	210	21	0.004952	0.495%
2026†	298	27.1	0.006381	0.638%

All values computed live from $m = 3$, $\sigma = 6$. †2026 values annualised from 103 days of data (28 documented attacks × 365/103). For reference, institutional gold storage costs 0.5–1.5%/yr. At the current 2026 pace, the BTC security cost (0.64%) has entered the gold-storage range. The 2022 spike reflects the FTX-era whale exodus (smaller denominator), not an attack surge.

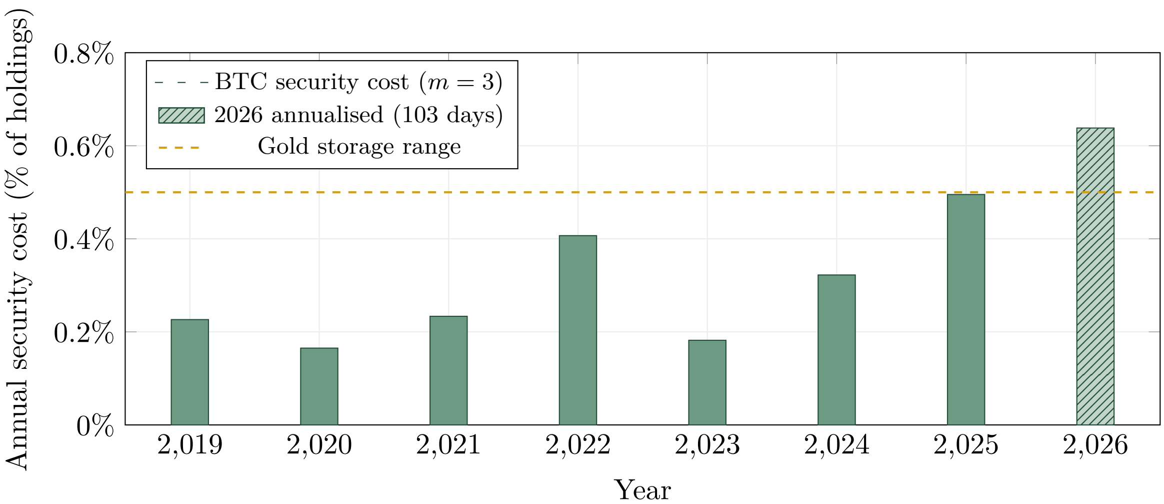

Figure 2: Insurance-equivalent annual security cost for individual Bitcoin whales (green bars), compared to the institutional gold-storage cost range (gold dashed: 0.5–1.5%/yr). The hatched 2026 bar is annualised from 103 days of data (28 attacks through April 13). At the current pace, the BTC security cost has entered the gold-storage range.

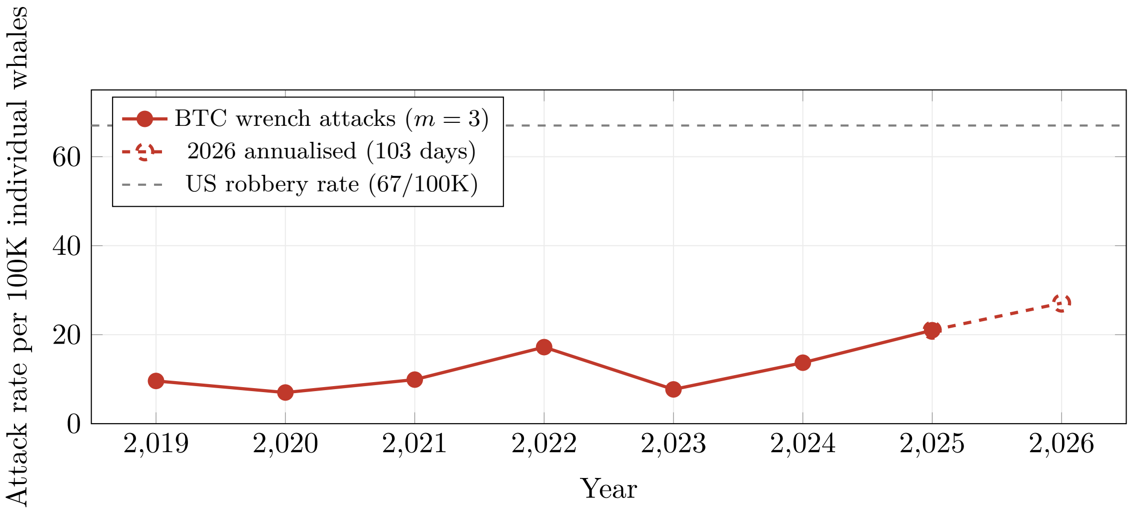

Figure 3: Per-capita wrench-attack rate for individual Bitcoin whales (at 3× underreporting), compared to the US general robbery rate. The open circle for 2026 is annualised from 103 days of data. The 2026 pace (27/100K) represents a 29% change relative to 2025.

Why the Trend Worsens

The current security cost (0.64%/yr) understates the trajectory for several structural reasons:

1. **Target identification is becoming easier.** Bitcoin is still held by far fewer people than jewelry, and typically more discreetly. Identifying a wrench attack target today requires blockchain analysis, social media intelligence, or insider knowledge — skills less diffused than identifying a jewelry owner (visible watches, known stores, insured collections). But as adoption grows, Bitcoin wealth becomes statistically correlated with observable proxies (neighborhood, profession, social media). The targeting problem is converging toward the jewelry baseline.

2. **Criminal skill sets are diffusing.** In 2014, a Bitcoin wrench attack required knowledge of wallets, private keys, and transfer mechanics. Today, a smartphone unlock and a Coinbase login suffice. The skill barrier is falling toward that of ordinary robbery.

   Obscurity-based defenses are a second layer of this same phenomenon — and knowledge diffusion causes them to *backfire*. Once an attacker learns what a decoy wallet is, they simply persist beyond the first transfer: the victim has complied, but the attacker has no way to know whether the disclosed wallet is the real one or a decoy. The rational response is to keep coercing. If the victim "protects" a UTXO via a two-path lock script — path A (clear key + n-block delay) vs. path B (obscure redirect key) — the victim complies using path A and hopes the attacker is unaware of path B, which can claw back the transfer during the delay window. An educated attacker who knows the scheme either coerces the victim to use the redirect key up front, or — for good measure — kills the victim after the coerced transfer to prevent them from redirecting it. In short, the efficacy of obscurity-based security is contingent on the premise that attackers are unable to acquire the same practical knowledge that defenders possess — knowledge that, in Bitcoin's case, is open-source by design. As Kerckhoffs articulated in 1883: a system's security must not depend on the secrecy of its mechanism.[16] Obscurity violates this principle structurally.

3. **The $100K threshold is falling in BTC terms.** In 2014, $100K required 314 BTC (early miners only). In 2025, it requires 1.1 BTC. The whale population is growing structurally as BTC appreciates, expanding the target set even without new adoption.

[16] Kerckhoffs, A., "La cryptographie militaire," *Journal des sciences militaires*, 1883.

4. **Attack CAGR exceeds whale CAGR.** Over 2019–2026, documented attacks grew at 43% per year while the individual whale population grew at 24%. The per-capita rate is increasing at a net 16% annually.

We are, in short, **early in the trend**. The current attack rate per whale (27/100K) is still well below mature crime categories like jewelry theft (∼50–200/100K). As the targeting and skill barriers fall, convergence toward those rates is the natural trajectory.

Forward-Looking Pricing

Why current cost understates the discount

A significant component of Bitcoin's value is speculative: priced on the expectation of future adoption and appreciation. Speculative assets are particularly sensitive to trend changes in holding costs. The analogy is real estate: if a neighborhood has strong fundamentals (good schools, growing employment) but a visibly rising crime rate, housing prices drop *before* the crime rate reaches high levels — because buyers price in the trajectory, not just the current level. No single crime incident causes the repricing; it is the *trend* that compounds into the discount.

Bitcoin exhibits the same structure. The market periodically attributes underperformance to rotating FUD cycles — China mining bans, regulatory crackdowns, quantum computing fears — each of which arrives, dominates a news cycle, and recedes. But the security tax does not recede. It compounds at 16%/yr and is priced by no market. A rational long-term holder internalizes this cost whether or not it appears in any headline. To return to the analogy: the neighborhood's "China ban" scare comes and goes, but the rising break-in rate persists across all cycles. The persistent component is the one that reprices the asset.

We do not claim the security tax *explains* Bitcoin's deviation from bullish models (stock-to-flow, etc.) — too many confounds exist. We claim it is a **structurally compounding holding cost that is currently priced at zero**, and that any rational pricing model should subtract it. The attached present-value calculation quantifies how large that subtraction is.

Estimating the growth rate g

The net growth rate of the per-capita attack rate is the difference between the attack growth rate and the whale population growth rate. We estimate each as a CAGR over 2019–2026 (7 years), using the annualized 2026 attack count (99):

$$
\begin{aligned}
g_{\text{attack}} &= \left(\frac{a^*_{2026}}{a_{2019}}\right)^{1/7} - 1 = \left(\frac{99}{8}\right)^{1/7} - 1 = 43.3\%/\text{yr} \\
g_{\text{whale}} &= \left(\frac{W_{2026}}{W_{2019}}\right)^{1/7} - 1 = \left(\frac{1100}{250}\right)^{1/7} - 1 = 23.6\%/\text{yr} \\
g_{\text{net}} &= \frac{1 + g_{\text{attack}}}{1 + g_{\text{whale}}} - 1 = 16\%/\text{yr}
\end{aligned}
$$

(a^*_{2026} = annualized from 28 attacks in 103 days.) The attack count is growing faster than the whale population, so the per-capita rate is rising at 16%/yr. We exclude pre-2019 because whale populations below ∼250K make per-capita rates unreliable (see Section 8).

Present value of expected security costs

If the per-capita attack rate continues growing at g = 16%/yr, the present value of security costs over a holding horizon H is:

$$D(t, H) = \sum_{k=0}^{H-1} \frac{c(t) \cdot (1+g)^k}{(1+r)^k} = c(t) \cdot \frac{q^H - 1}{q - 1}, \qquad q = \frac{1+g}{1+r} \tag{3}$$

The sum walks forward through each year of the holding period. In year k, the security cost has grown to $c(t) \cdot (1+g)^k$ — because both the absolute attack count and the per-capita rate are increasing — but that future cost is worth less today by the discount factor $(1+r)^{-k}$, where r is the holder's required return. The ratio $q = (1+g)/(1+r)$ captures the race between cost growth and discounting:

- When $g < r$: future costs grow, but discounting shrinks them faster. The series converges. Each additional year contributes less.
- When $g > r$: costs outrun discounting. Each additional year of holding contributes *more* than the last, and the sum diverges for $H \to \infty$. This is the regime implied by the current data ($g \approx 16\%$, likely exceeding most real required returns).

$D(t, H)$ is the fraction of current holdings that a rational buyer should subtract from the asset's fundamental value to compensate for expected security costs over the next H years. It is the present value of the stream of annual "insurance premiums" that no insurer offers.

Horizon	$r = 10\%$	$r = 15\%$
5 years	3.56%	3.24%
10 years	8.18%	6.63%

At a 10% required return and 10-year horizon, the forward-looking discount is 8.2% — roughly 13× the current annual cost. This amplification occurs because the market must price in *all* expected future security costs, not just this year's.

Caveat: the 16% net growth rate will not persist indefinitely — either security solutions emerge (capping the cost) or adoption slows (reducing new whales). The finite-horizon formula is an upper bound under the assumption that the current trend continues. This is precisely the window in which intervention matters most.

Comparison to Gold

Gold is the natural comparator: a portable store of value that faced the same physical vulnerability problem over centuries.

	Gold (vaulted)	**BTC (self-custody)**
Annual holding cost	0.5–1.5%[17]	0.64% (and rising)
Insurance available	Yes (mature market)	No
Loss per event	Partial (∼30%)	Total (100%)
Cost trend	Stable	Worsening (+16%/yr)
Security infrastructure	Vaults, Brinks, Fort Knox	"Don't tell anyone"

Gold's explicit storage/insurance costs are *known, stable, and priced.* Bitcoin's implicit security costs are *uncertain, growing, and unpriced* — exactly the conditions under which risk-averse capital rotates away.

The concurrent gold surge (from $1,393/oz in 2019 to $3,435/oz in 2025, +147%) is consistent with this rotation. Capital that might otherwise flow into BTC as a store of value flows instead into an asset whose security costs are solved, explicit, and declining in real terms.

Robustness

Sensitivity to underreporting multiplier

Multiplier m	2026 rate/100K	2026 cost	10-yr PV (r=10%)
1× (reported only)	9	0.213%	2.73%
3× (Lopp estimate)	27.1	0.638%	8.18%
10× (aggressive)	90.2	2.127%	27.28%

The underreporting multiplier scales both the annual cost and the PV linearly. Even at 1× (reported attacks only), the 10-year forward-looking discount is non-trivial.

Why $m = 3$ is conservative. Jewelry theft provides a floor for calibrating underreporting, because it is the closest analog (portable, high-value, personally held) with established victimization survey data. The NCVS–UCR gap for property crime implies that only ∼30–33% of property crimes are reported to police.[18] Reporting rates vary sharply with insurance status:

Jewelry victim profile	Est. reporting rate	Implied m
Insured, declared	50–65%	∼1.5–2×
Uninsured, nothing to hide	25–35%	∼3–4×
Undeclared / off-book holdings	5–15%	∼7–20×

Insurance is the dominant driver of reporting: a police reference number is required to file a claim.[19] Over 60% of Americans carry no jewelry insurance.[20] For undeclared assets, reporting the theft means documenting holdings the victim never disclosed — creating exposure for tax evasion, customs violations, or unreported inheritance. The FINRA Foundation finds that only ∼15% of financial fraud victims report, with shame and self-blame as the primary barriers.[21]

Bitcoin wrench attacks compound every factor that suppresses jewelry reporting, then add more:

- **No insurance exists** — the single largest driver of property-crime reporting is entirely absent.
- **Recovery rate is zero** — transferred BTC is irrecoverable; there is no property to recover.
- **Cultural norm is non-disclosure** — the community's own security advice ("don't tell anyone you own Bitcoin") primes victims against reporting.
- **Undeclared holdings are prevalent** — BTC held for tax optimization, sanction avoidance, or simply outside the regulated financial system creates an active *disincentive* to involve authorities. Reporting the attack means documenting assets the victim never disclosed.
- **Victim-blaming compounds shame** — "you should have used cold storage" / "you shouldn't have told anyone" mirrors the shame dynamic that suppresses fraud reporting to ∼15%.

If *insured* jewelry — the most-reported category of portable-valuables theft — already implies $m \approx 1.5$–2, and *uninsured off-book* jewelry reaches $m \approx 7$–20, then Bitcoin — which shares every underreporting factor with jewelry and adds several more — should sit well above the jewelry floor. The $m = 3$ baseline is better understood as the *lower bound of the jewelry analog*, not a Bitcoin-specific estimate. A range of $m = 5$–10 is defensible; the $m = 10$ row in the table above may be closer to ground truth than the highlighted baseline.

[18] BJS, *Criminal Victimization, 2022*; see also Skogan, W.G., "Dimensions of the Dark Figure of Unreported Crime," *Crime & Delinquency* 23(1), 1977.

[19] BJS, *Reporting Crimes to the Police* (Harlow, 1985): for losses ≥\$250, "economic incentives (to collect insurance or recover property) dominated."

[20] ValuePenguin, "More Than 6 in 10 Say Their Jewelry Is Uninsured," 2023.

[21] FINRA Foundation, *Blame and Shame in the Context of Financial Fraud*, 2021.

Sensitivity to severity multiplier

σ	Interpretation	2026 cost	10-yr PV (r=10%)
3	Total loss only; no uninsured premium	0.319%	4.09%
6	Baseline (total loss × uninsured)	0.638%	8.18%
10	Including violence/trauma premium	1.064%	13.64%
15	Full irrecoverability + behavioral premium	1.595%	20.46%

At $\sigma = 15$ (defensible if behavioral/psychological costs are included), the 10-year discount reaches 20.5%.

Sensitivity to whale population

The whale population estimates are the least certain input. We show the 2026 security cost at three population levels:

Whale pop. (2026)	Rate/100K	Ann. cost	10-yr PV (r=10%)
500K (conservative)	59.5	1.404%	18%
1 100K (baseline)	27.1	0.638%	8.18%
2 000K (generous)	14.9	0.351%	4.5%

Even with 2M whales (generous upper bound), the 10-year discount remains material.

Limitations

1. **Whale population estimates are rough.** On-chain data conflates addresses with individuals, and institutional vs. personal holdings are not cleanly separable. We mitigate this with sensitivity analysis (Section 8).
2. **The severity multiplier is a judgment call.** The $\sigma = 6$ baseline is conservative and grounded (total-loss ratio × uninsured premium), but the true behavioral discount could be higher or lower.
3. **Underreporting is uncertain.** Lopp estimates 2–3×; law enforcement suggests it could be higher. The model is linear in m.
4. **The insurance analogy is approximate.** Jewelry insurance pricing reflects a mature market with actuarial data, loss adjusters, and reinsurance. Applying the same per-unit-of-crime pricing to Bitcoin is a first-order transfer, not an exact calibration.
5. **Growth rates will not persist.** The 16% net growth rate of per-capita attacks implies convergence toward mature crime rates within a decade. The model assumes constant growth over the horizon; reality will involve plateaus and discontinuities.
6. **No behavioral/media amplification factor.** The model prices actuarial risk only. The "SF Union Square effect" — where anti-crime measures and media coverage suppress demand far beyond the actuarial loss — is real but not quantified here. The insurance-based estimate is therefore a *lower bound*.

Testable Predictions

1. **Insurance market emergence.** If the security cost estimate is correct, an insurance market for self-custody BTC should emerge with premiums in the 0.5–2% range. Lloyd's of London or specialty crypto insurers are the likely first movers.

2. **Luxury-goods crime correlation.** The geographic distribution of wrench attacks (per Lopp's tracker) should correlate with luxury-goods crime indices. France's outsized share of attacks should correlate with France's jewelry/watch robbery rates.

3. **Whale migration to custodial solutions.** As the security cost rises, the fraction of whale-held BTC in self-custody should *decrease* in favor of institutional custody and ETFs, despite the community's philosophical preference for self-sovereignty.

4. **The 2026 trajectory — tracking the model.** The 2025-based projection implied ~90 attacks for 2026 on a trend-line extrapolation. As of April 13, 2026, 28 attacks have been documented in 103 days — an annualised pace of 99, already above the full-year 2025 total and within 10% of the extrapolation. At that pace, the 2026 annual security cost reaches 0.638%, firmly inside the gold-storage range. The acceleration remains concentrated in France (20 of 28 YTD attacks), where data breaches have created target-rich conditions.

5. **Great Wall adoption should suppress the cost trajectory.** If coercion-resistant custody reaches meaningful share among individual whales, the effective severity multiplier should decline (both loss probability and loss magnitude fall), providing a natural experiment.

What Great Wall Does About It

The security tax reframes Great Wall's value proposition in concrete, defensible terms:

Great Wall eliminates an implicit annual holding cost that already sits inside the gold-storage range — without requiring obscurity, custodial trust, or a self-sovereignty compromise.

Great Wall is a free-software self-custody Bitcoin wallet, dual-licensed under MIT or Apache-2.0, in which the seed lives only in the user's memory. The authentication secret is encoded onto a user-specific perturbation of the Burning Ship fractal; recall is spatial-visual, and — as elaborated below — the holder *cannot articulate* the secret in a form an attacker could transcribe. The entry point is gated by a calibrated Argon2 derivation whose target duration is benchmarked against Jameson Lopp's published wrench-attack corpus, so that no published custody window would have sufficed to extract the secret.

Scale matters for the argument. At the current annualised cost of 0.64%, a whale population of ~1.1M individuals, and an average of ~\$300K per whale in BTC, the aggregate annual security tax is approximately:

$$0.64\% \times 1{,}100{,}000 \times \$300{,}000 = \$2106\text{M/year}$$

Over a 10-year horizon with the forward-looking multiplier, the present value of the security tax across the whale population is 8.2% of \$330B ≈ \$27B. That figure is value being destroyed across the entire Bitcoin ecosystem, not a revenue projection. The engineering path through which Great Wall eliminates it — fractal encoder, time-lock-puzzle library, spaced-repetition training vault, Lightning marketplace for outsourced solving, dead-man's-switch inheritance protocol, and the unified desktop wallet that integrates them — is the subject of the sections that follow.

The four-property test: why every existing custody method fails

Before the engineering details, it is worth stating what an adequate self-custody protocol must do. The requirement is the simultaneous satisfaction of four security properties that, until now, have been mutually exclusive across deployed custody methods:

- **Knowledge-Based (Deviceless).** Access is tied only to the user's mind — to *tacit knowledge* — not to a physical device, seed phrase, or object that can be stolen, lost, or seized. A device that exists in the world is a device that can be taken from it.

- **Individual Custody.** The protocol upholds the core cryptographic promise: *not your keys, not your coins*. The holder is their own bank; no third-party custodian, exchange, or co-signer stands between the holder and their funds.
- **Anti-Obscure (Anti-Fragile).** Obscurity is fragile (Section 5): once attackers learn the trick, the protection collapses and often inverts. An anti-obscure protocol is *anti-fragile*: its security *increases* as adversaries learn the mechanism, because public knowledge that the protocol cannot be coerced is itself the deterrent against attempting coercion.
- **Coercion-Resistant.** The architecture — requiring both *time* and *tacit knowledge* — makes it effectively impossible for an attacker to obtain access through force within any window the attacker can sustain. The $5 wrench is neutralised at the protocol layer, not at the operational-security layer.

Every other custody method on the market satisfies a strict subset of these four properties — and the property it fails on is, in each case, the one that gets the holder killed, robbed, or expropriated. The table below summarises the comparison.

Method	Dev.	Indiv.	Anti-Obsc.	Coerc.	Examples	Primary down-side
TKBA	✔	✔	✔	✔	Great Wall.	relies on the user not forgetting their tacit knowledge (**which we solve** via an integrated memory coach).
Physical	✘	✔	✔	✔	physical vault(s), multiple addresses, Shamir's secret sharing / multisig	**astronomically high cost** and operational complexity, and geographic binding: "glorified gold" (see Section 5).
Shared / Delegated	✔	✘	✔	✔	exchanges or co-custody companies: Coinbase, Binance, Casa.	**negates the core premise** of self-custody: "not your keys. . . "
Obscurity-Based	✔	✔	✘	☠	decoy wallets; plausible denial; redirectable time-locked transactions	**inevitably backfires**: once educated about the obscure method, the attacker has a material incentive towards **more violence** (torture / assassination).
Vanilla Custody	✔	✔	✔	✘	hardware (Ledger, Trezor, Krux BIP39) or software (Electrum, Exodus, MetaMask) wallets	**completely vulnerable** to a wrench attack.

Read column-by-column, the table is a statement about the structure of the custody-design space: the four properties are not a wish-list but a *covering set*. Drop any one of them and the failure mode is concrete and named. Physical schemes drop **Deviceless** and inherit the cost/geography burden that Bitcoin was created to escape. Shared/delegated schemes drop **Individual Custody** and inherit the very counterparty risk the Bitcoin thesis exists to retire. Obscurity-based schemes drop **Anti-Obscure** and convert robbery into homicide as the attacker's knowledge of the trick grows — the red skull in the coercion column denotes *negative* resistance, not merely absence. The mechanism is no longer hypothetical. Once the community's own consensus advice is "deny that you own Bitcoin" and "hand over a decoy if pressed," *every signal a victim can emit under coercion has been pre-discredited by the defence itself*: a denial is presumed to be the denial-script; a disclosed wallet is presumed to be the decoy. An educated attacker therefore has no Bayesian update available short of continued violence, and the rational continuation of an interrogation that began with a denial is more violence, not less. A documented case fits the prediction with grim precision: a Swedish League of Legends streamer was bound and tortured for hours by armed intruders demanding cryptocurrency he allegedly did not own; the attackers — evidently unable to credit the denial — persisted long past any actuarial value the robbery could have justified, ultimately leaving with the victim's Pokémon-card collection.[22] The victim's profile (a publicly visible high-income gamer) supplied the prior; the obscurity norm supplied the discount on his denial; the attackers supplied the rest. Vanilla wallets drop **Coercion Resistance** and reduce the user's last line of defence to silence about their holdings — exactly the maximalism-vs-obscurity contradiction with which this document opened.

The case repays closer reading, for its interpretation *forks* — and every branch indicts the obscurity strategy. **(i)** If the denial of holdings was itself a defensive lie, the episode is a textbook obscurity failure: secrecy did not avert the attack, and the victim's only residual recourse is to *double down* on the denial even after the assault, wagering against a repeat — a wager made worse by the visibility the case's own notoriety produced. **(ii)** If the denial was true, he bore obscurity's *negative externality* directly: a genuine non-holder, attacked only because actual holders are coached to look exactly like him — which is precisely what collapses the statistical distinction between holder and non-holder. **(iii)** If the truth lies in between — a real but smaller-than-presumed stash — both pathologies operate at once: the externality redirected the attack onto him, *and* obscurity failed to deter it anyway. **(iv)** The decisive point is that all three readings are simultaneously plausible. Whether or not it holds for this victim specifically, each outcome is — in the aggregate, across a growing population of holders and lookalikes — all but certainly already occurring and set to increase; and the prevailing obscurity-based culture, exactly as Kerckhoffs's principle predicts, has spectacularly failed to prevent any of them.

TKBA — memory-held entropy navigated against a user-specific perturbation of the Burning Ship fractal, gated by a calibrated Argon2 derivation, with a verifiable delay outsourced to an anonymous Lightning marketplace — is the first construction that holds all four columns simultaneously. The remaining sections describe how each component is built from standard primitives.

The protocol is application-agnostic

Great Wall's time-lock layer enforces a cryptographic delay on any secret — Bitcoin keys being the first and most natural use case, but any application requiring a verifiable off-chain delay fits the same primitive. The protocol is open and runs entirely locally; only the *serial computation* that imposes the time-lock is outsourced, to an anonymous Lightning-Network marketplace whose role is to sell compute, not custody.

Time-lock mechanics

Great Wall's time-lock is built on *repeated modular squaring*, the construction proposed by Rivest, Shamir, and Wagner (1996)[23] and refined by modern Verifiable Delay Functions.[24] The setup:

- Choose an RSA modulus $n = pq$ of b bits (e.g. $b = 2048$).
- To lock for a target delay d: set $T = d \times s_{\max}$, where $s_{\max}$ is the squarings-per-second of the fastest plausible hardware.
- Unlocking requires computing $x^{2^T} \bmod n$ — exactly T sequential squarings.
- **Critical property:** without the factorization of n, this computation *cannot be parallelized.* Ten thousand cores offer zero speedup. Only single-pipeline sequential throughput matters.

The three variables are the modulus size b, the iteration count T, and the hardware squaring rate $s(b, H)$ for hardware class H. The squaring rate scales as $s(b, H) \approx C(H)/b^{\alpha}$, where $\alpha \approx 2$

[22] Lopp, J., post on X reporting the incident: "Swedish League of Legends streamer RATIRL reveals that armed intruders broke into his home, tortured him for hours, and demanded cryptocurrency he didn't have. The robbers eventually left with only his Pokémon cards. 2 suspects have been arrested thus far." `https://x.com/lopp/status/2058917155530293600`.

[23] Rivest, R.L., Shamir, A. & Wagner, D.A., "Time-Lock Puzzles and Timed-Release Crypto," MIT/LCS/TR-684, 1996.

[24] Boneh, D. et al., "Verifiable Delay Functions," *CRYPTO 2018*, LNCS 10991; Wesolowski, B., "Efficient Verifiable Delay Functions," *EUROCRYPT 2019*, LNCS 11478; Pietrzak, K., "Simple Verifiable Delay Functions," *ITCS 2019*, LIPIcs 124.

(Montgomery multiplication[25]) and $C(H)$ captures clock speed, IPC, and algorithmic tuning.

The **convenience ratio** $R = C(H_{\max})/C(H_{\text{user}})$ determines how many multiples of the minimum delay a given client experiences. Crucially, R is **independent of** b: the modulus cancels in the ratio. Choosing a larger modulus makes every tier proportionally slower but does not change the relative speed spread. The modulus should be chosen for cryptographic soundness (2048+ bits), not as a pricing lever.

Because the computation is inherently sequential, the speed spread across *all available hardware* is compressed to single-pipeline performance differences — roughly 10–30$\times$ from consumer laptop to hypothetical ASIC. This is simultaneously a **feature** (the attacker's advantage over any client is bounded by this narrow band) and a **constraint** (limited room for tier differentiation).

The guaranteed delay is hardware-contingent: $d = T/s_{\max}$ holds as long as $s_{\max}$ reflects the state of the art. In practice this is bounded: Moore's law yields $\sim$1.1–1.2$\times$/yr for single-thread performance, and the user re-calibrates T at each vault refresh cycle ($\sim$weekly). A VDF construction adds efficient verifiability — the solver returns a short proof of correct computation, removing the need to trust the solver.

Why weekly: the TKBA refresh cycle. Great Wall's coercion resistance rests on *Tacit Knowledge-Based Authentication* (TKBA) — authentication by navigating to a randomly generated identifiable location in a fractal landscape. Because all reference points in a fractal are self-similar, abstract, and jagged, the location is nearly impossible to verbally describe or sketch — even under duress. The holder recognises and navigates to their point, but cannot *communicate* it — Polanyi's "we can know more than we can tell."[26] This spatial memory is subject to *catastrophic forgetting* if not periodically rehearsed.[27] The vault refresh cycle ($\sim$weekly) is therefore not arbitrary: it is the cadence at which the holder must *revisit* their fractal location to maintain reliable recall. This same cycle conveniently re-calibrates T against the current computational state of the art and generates a steady stream of puzzles for the anonymity set.

Transport: Lightning Network + Tor

The marketplace runs natively over the **Lightning Network**. Puzzles, solutions, and payments all flow over the same LN channels, using onion-routed payments. Combined with Tor (already supported by LN wallets such as Phoenix and Zeus), the puzzle submission is unlinkable to the submitter. This solves three problems simultaneously:

- **Payment privacy.** Per-puzzle fees are micropayments — exactly what LN is designed for. No on-chain footprint per transaction.
- **Puzzle privacy.** The puzzle and its solution travel over the same onion-routed channel as the payment. No separate transport protocol needed.
- **Infrastructure perk.** Both clients and solvers become well-connected LN nodes. If the marketplace achieves scale, its channel graph becomes significant LN routing infrastructure. Participants gain high-connectivity wallets useful for *all* LN payments, not just GW — creating stickiness beyond the anonymity set.

A note on the marketplace posture

Whatever fees the marketplace ultimately charges must be *value-blind*: holders do not want to disclose how much (or what) their time-lock protects, so any AUM-based or value-proportional scheme would leak the very information the scheme is designed to protect. A flat per-puzzle fee is the only posture consistent with the privacy property. This document does not propose a fee schedule; it notes only that the privacy constraint rules out any fee that is a function of the protected

[25] Montgomery, P.L., "Modular Multiplication Without Trial Division," *Math. of Computation* 44(170), 1985.
[26] Polanyi, M., *The Tacit Dimension*, Routledge & Kegan Paul, 1966.
[27] Ebbinghaus, H., *Über das Gedächtnis*, 1885; Murre, J.M.J. & Dros, J., "Replication and Analysis of Ebbinghaus' Forgetting Curve," *PLOS ONE* 10(7), 2015.

value, and that the anonymity set itself — the pool of concurrent puzzles among which any individual puzzle is indistinguishable — is the core property the marketplace has to preserve.

Inheritance as a dead-man's switch

The same time-lock infrastructure underwrites a trustless inheritance protocol. Its role in this document is technical — it shows that one building block, the RSW time-lock, is doing double duty for coercion resistance and for post-mortem transfer — rather than commercial.

Setup. A testator (Ted) holds BTC across LN channels with GW marketplace nodes and wishes to distribute portions $p_1, p_2, \ldots$ to heirs Henry, Harry, etc.

1. Ted creates an HTLC (Hash Time-Locked Contract) in his GW channel: "Pay p_1 to Henry's key if the preimage of hash H is revealed."
2. A time-lock puzzle, when solved, reveals the preimage.
3. Ted broadcasts the puzzle to Henry via LN (onion-routed, private).
4. **Rotation:** Before the puzzle is solved, Ted cancels the old HTLC and creates a new one with a fresh hash — both are channel state updates, *entirely off-chain.* The old puzzle, even if eventually solved, reveals a preimage for an HTLC that no longer exists.
5. While Ted is alive and competent, he rotates periodically (∼weekly). Each rotation invalidates the previous puzzle and issues a new one.
6. **Trigger:** When Ted dies or becomes incapacitated, rotation stops. The current puzzle runs to completion. Henry solves it, reveals the preimage, and claims the HTLC.

Key properties:

- **No trusted third party.** No lawyer, no custodian, no multisig co-signer. The cryptographic time-lock *is* the dead man's switch.
- **No on-chain footprint during Ted's lifetime.** HTLC creation, cancellation, and rotation are channel state updates — off-chain. Ted's BTC stays in existing LN channels; rotation moves balances between channels via circular LN payments (also off-chain). The only on-chain transaction occurs when an heir force-closes a channel to claim inheritance — after Ted's death, when privacy no longer matters.
- **Proportional distribution.** Different heirs receive different portions, each with its own HTLC and puzzle stream.
- **Automatic trigger.** No "death oracle" or certificate needed. Cessation of rotation *is* the trigger.

Anonymity-set contribution. A testator with three heirs rotating weekly issues $3 \times 52 = 156$ puzzles per year. These outstanding (unsolved) puzzles pad the anonymity set for anti-coercion users — more concurrent puzzles in the pool means better privacy for everyone — but are not themselves solved until rotation stops. Inheritance and anti-coercion are therefore mutually reinforcing at the protocol level: each increases the other's anonymity-set depth without requiring coordination.

Why This Is Bitcoin-Ecosystem Infrastructure

The argument of this document is simple: the Bitcoin community has a measurable, compounding physical-security cost that it currently prices at zero, that its consensus defence (obscurity) fails to address and indeed contradicts in its own terms, and that a purely cryptographic alternative exists and is implementable from existing primitives. The Great Wall deliverables — fractal encoder, time-lock-puzzle library, spaced-repetition training vault, anonymous Lightning marketplace for outsourced solving, dead-man's-switch inheritance protocol, and a unified desktop wallet integrating the six libraries — are the engineering path from the pricing argument to a working system.

The project's posture is that of ecosystem infrastructure:

1. **Every wrench attack is a news cycle reinforcing that self-custody is unsafe, and every "don't tell anyone you own Bitcoin" is a brake on adoption.** The security tax

quantified above — \$2106M per year and growing at 16%/yr — is value being destroyed across the entire Bitcoin ecosystem. A working TKBA wallet eliminates it.

2. **Standard Bitcoin primitives only; no soft fork ask.** Taproot outputs, MuSig2 aggregation, standard Lightning commitment/revocation, BIP39/BIP32 derivation, secp256k1. The time-lock lives one layer above the script, wrapping key material rather than adding a new opcode.
3. **Reusable building blocks.** The RSW time-lock library, the anonymous LN marketplace client, and the dead-man's-switch channel pattern are independently useful for any other Bitcoin project that needs long-horizon conditional spending without covenants.
4. **Purely free software.** All seven repositories are dual-licensed MIT or Apache-2.0. There is no equity, no token, no pre-sale, no valuation overlay. A prior commercially-structured attempt at the same core idea was wound down; the current effort is a clean-slate restart into the grant-funded open-source posture described here.

The return is primarily an *externality*: a healthier Bitcoin ecosystem benefits every participant. This document makes the analytical case for why that software is worth producing; the engineering time required to produce it is the subject of the Great Wall repositories referenced in the footer.

The insurance-based framework provides a lower bound on the security cost (actuarial only, no behavioural amplification). All values are computed live by LaTeX from the parameters in the preamble; changing any assumption recomputes the entire analysis.

Great Wall umbrella repository: `https://github.com/Yuri-SVB/Great-Wallet`.

Attack data: `https://github.com/jlopp/physical-bitcoin-attacks` *(refreshed 22 April 2026; 2026 figures above reflect the Jan–Apr 13 window in that tracker).*